ALICE L WALTON SCHOOL OF MEDICINE

Pharmacodynamics2: Quantal Dose Response and Therapeutic Index

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Adapted from: Pharmacology: Foundations and Frameworks

Listen to this Brick

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Learning Objectives (3) | Versions

After completing this brick, you will be able to:

1. Explain distinct mechanisms of action of chemical and physiologic antagonists and identify their roles within specific clinical applications.
2. Interpret qu g attributes such as TD50, ED50, safety, and determine the therapeutic index.
3. Identify relevant pharmacodynamic and pharmacokinetic principles and apply those to solve clinical problems and predict outcomes.

CASE CONNECTION

RT is a 40-year-old female patient who came to your outpatient clinic. She is currently on warfarin (anticoagulant) (5mg once daily), and her PCP has recommended that she visit the outpatient anti-coagulant clinic at the hospital. RT has come in for her first visit, and she just found out that she is scheduled for a weekly appointment for the next three weeks. RT is frustrated with the frequency of her appointments. "Why do I have to come in weekly for this blood thinner? I do not have to do that with my other medications." She also says, "I don't like this. Can I stop taking it?" When RT is asked about her concerns, she repeats her frustration regarding the weekly appointments. RT states, "I have a job, and it's hard to take time out every week for a visit." As a resident, you are in the exam room with the attending. You tell RT that medication and weekly monitoring are equally important.

How can you explain RT the importance of weekly monitoring of warfarin to RT? Consider your answers as you read, and we'll revisit RT at the end of the brick.

GO TO CONCLUSION ↓

Are there other types of antagonism? (LO#1)

Many drugs bind dire ceptor, stimulating or inhibiting the action of that enzyme or receptor, such as drugs that bind to dopamine receptors. In the Pharmacodynamics-1 brick, we discussed agonist and antagonist interactions at a single receptor level. However, some antagonist interactions do not involve the receptor at all.

Physiologic Antagonism (also known as functional antagonism) occurs when two drugs reduce the activity of an enzyme- or receptor-activated process by producing the *opposite physiologic effect (usually acting on different physiological systems)*. These effects are typically compensatory mechanisms that are useful in maintaining homeostasis. For example, glucagon is considered a physiologic inhibitor of insulin. Insulin and glucagon are endogenous hormones that play an important role in glucose metabolism and uptake. Insulin signals the body to store glucose from the blood as glycogen, and glucagon has the opposite effect. Glucagon signals the body to break down glycogen stores and release glucose into the blood.

CLINICAL CORRELATION

Glucagon is an endogenous hormone, but it can also be used as a drug to treat patients with severe hypoglycemia (low blood sugar), such as a patient with diabetes who used too much insulin.

Chemical Antagonism occurs when one drug interacts with and inactivates another drug by a *specific chemical reaction* apart from enzyme inhibition. Anticancer drugs (eg, cisplatin) are chemical inhibitors that intercalate into DNA to cause damag e another good example of chemical inhibitors. Many antacids (eg, sodium bicarbonate, magnesium hydroxide, and aluminum hydroxide) are basic compounds that bind to stomach acid to neutralize it.

What is an example of a physiologic inhibitor?

How Do We Measure Drug Effects? (LO#2)

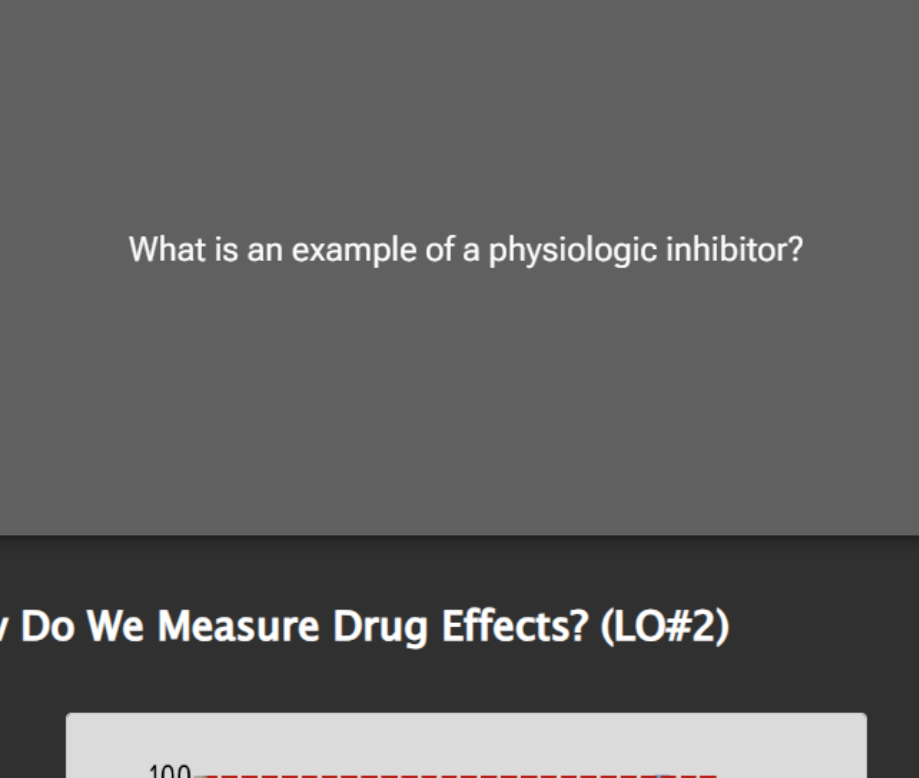

Figure 1

In the Pharmacodynamics1 brick, you have seen a graded dose-response curve in Figure 1. In this relationship, the drug effect increases in proportion to increasing drug dose shows increasing drug concentrations for two drugs. This plot was also used to determine the Emax and EC$_{50}$ for the drug, which is the concentration producing 50% of the maximal effect. The graded dose response curves can also help us determine the potency, efficacy, and affinity for a drug. The activity below should help you recall those concepts. (Note: The EC$_{50}$ should not be confused with ED$_{50}$ which is discussed later.)

ACTIVE LEARNING QUESTION

Match the term to its correct definition

potency	ven effect
efficacy	ability to produce max response
affinity	how well a drug binds to its receptor

Explanation

Potency: the amount of drug required to produce an effect of intensity. Differences in drug potency are evaluated by comparing EC50 (or ED50) values. **Efficacy**: the ability of a drug to produce a maximum response. Differences in drug efficacy are evaluated by comparing differences in maximal response at high drug doses or concentrations. **Affinity** can be defined as the tenacity (or how well) with which a drug binds to its receptor. Affinity is how well the drug binds to the receptor, while efficacy is what happens after the drug is bound to the receptor.

What is the difference between potency and efficacy?

In comparison, when we describe administration of a drug to the patient, we refer to the 'concentration' as a dose. The reason is that when given to a patient, the 'concentration' achieved in the body will depend on multiple factors such as distribution or metabolism. Therefore, we usually refer to the concentration as a dose.

A graded dose-response curve is concerned with plasma drug concentration for a single patient. However, we are often looking for information for dosing in a given population. **Quantal dose-response curves** are used to examine the effect of increasing concentrations of a drug on a population (Figure 2). Graded dose-response graphs plot response as a function of drug concentration, whereas quantal dose-response curves describe the proportion of a population exhibiting a defined outcome at a given dose.

What does a quantal dose-response curve represent?

The Quantal dose-response curves help determine another important pharmacodynamic parameter, **median effective dose (ED) or ED$_{50}$**. If we examine Figure 2, on the x-axis is the drug dose (in mg/kg) and y-axis is the number of patients responding to the drug (for example headache, changes in blood pressure, etc.). If we plot the dose against the number of patients responding, a frequency curve is generated, which has a Gaussian distribution (i.e., bell curve). One way to define ED$_{50}$ is the dose at which 50% of the subjects (population) respond.

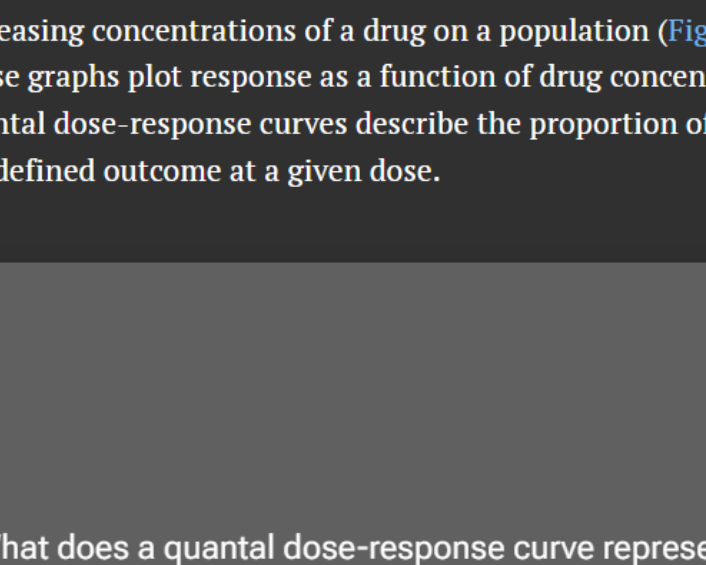

Figure 2

The percent populati s, and the relationship is quantal which means that either the effect is present or absent in a given individual. The response cannot be quantified; it is either present or absent. In other words, Quantal dose response studies are measuring all or none responses in large patient populations. For example, asking a question: does a drug x dose produce hiccups within 6 hours? The quantal dose response will help us determine the median dose for causing hiccups within 6 hours, i.e ED$_{50}$, but also help determine the minimum dose that could cause hiccups within 6 hours for drug x. Other examples of quantal events can be headache, cardiac arrest, or death from a drug reaction.

If the quantal responses shown in figure 2 were summated at a given dose on a log scale, then the resulting cumulative distribution would look like a sigmoidal dose response (Figure 3) curve. The curve represents the percentage of subjects that showed the drug effect. However, this is now a quantal dose-response curve (figure). The quantal dose-response curve can be used to generate the median effective dose (ED$_{50}$). The ED$_{50}$ is the dose at which 50% of the individuals show the specified quantal effect. The utility of knowing ED$_{50}$ is that it helps clinicians anticipate the drug effects and allows the drug manufacturer to recommend dose range for prescribing.

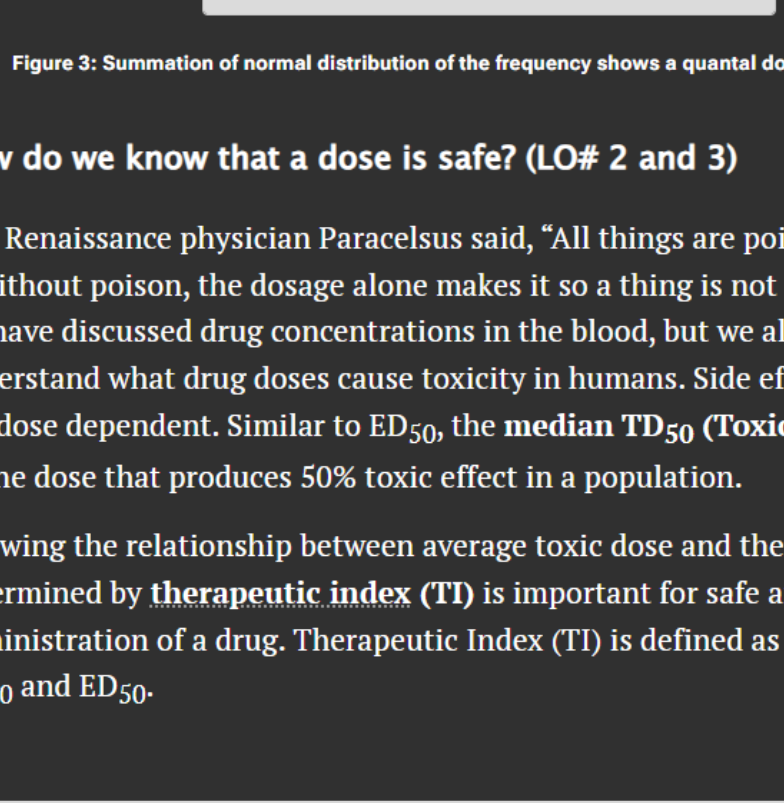

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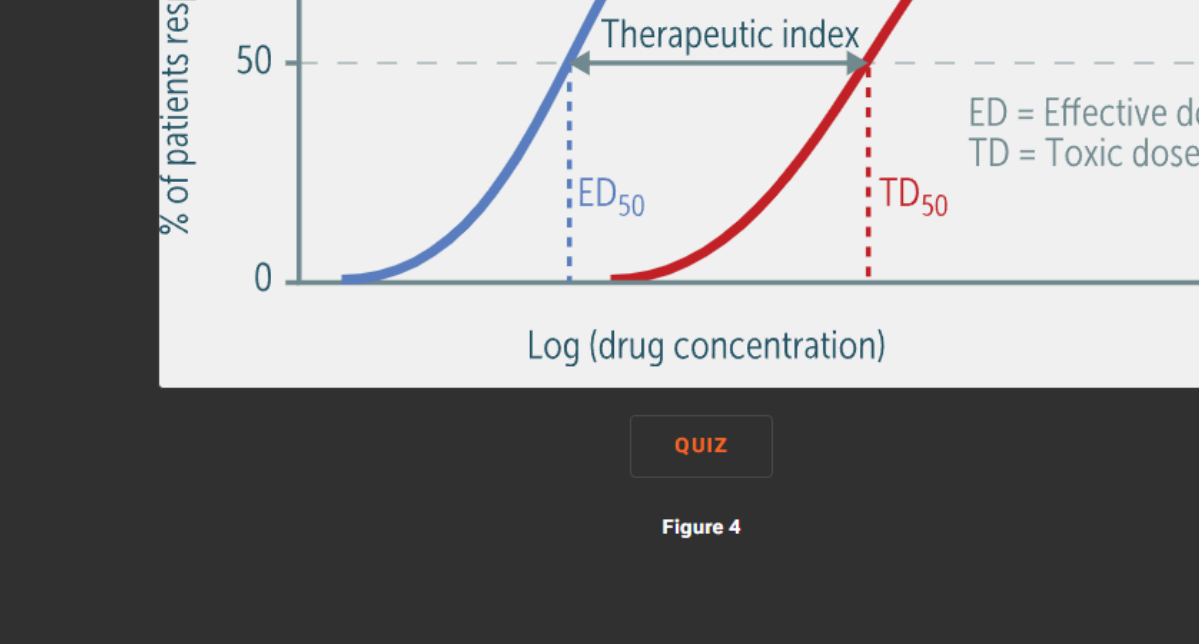
Figure 3: Summation of normal distribution of the frequency shows a quantal dose-response curve

How do we know that a dose is safe? (LO# 2 and 3)

The Renaissance physician Paracelsus said, "All things are poison, and nothing is without poison, the dosage alone makes it so a thing is not a poison." So far, we have discussed drug concentrations in the blood, but we also need to understand what drug doses cause toxicity in humans. Side effects from a drug are dose dependent. Similar to ED$_{50}$, the **median TD$_{50}$ (Toxic dose)** is defined as the dose that produces 50% toxic effect in a population.

Knowing the relationship between average toxic dose and the therapeutic dose, determined by **therapeutic index (TI)** is important for safe and effective administration of a drug. Therapeutic Index (TI) is defined as the ratio between TD$_{50}$ and ED$_{50}$.

$$TI = TD_{50} \div ED_{50}$$



QUIZ

Figure 4

From the curves below, you can see that the larger the TI, the lesser overlap between the efficacy and toxicity curves, the safer the drug. It helps determine the relative safety margin of a drug. Also, TI can help guide the required therapeutic level of monitoring. For example, monitoring of drugs is required for drugs with a narrow therapeutic index. In some cases, TI can be used to guide dosing intervals.

If the dose is increased beyond the toxic dose, then it could possibly be lethally harmful to the patient. Therefore, the **median lethal Dose (LD$_{50}$)** is defined as the dose required to (Figure 5).

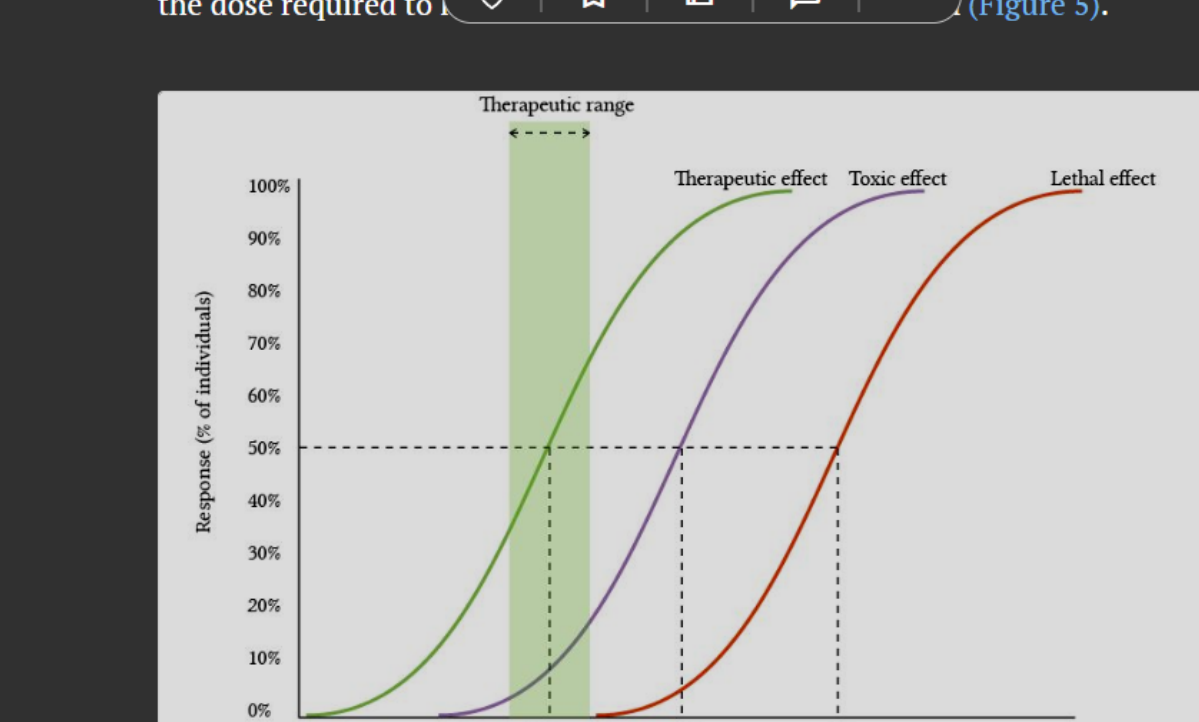


Figure 5: Dose response effect showing TD$_{50}$ and LD$_{50}$

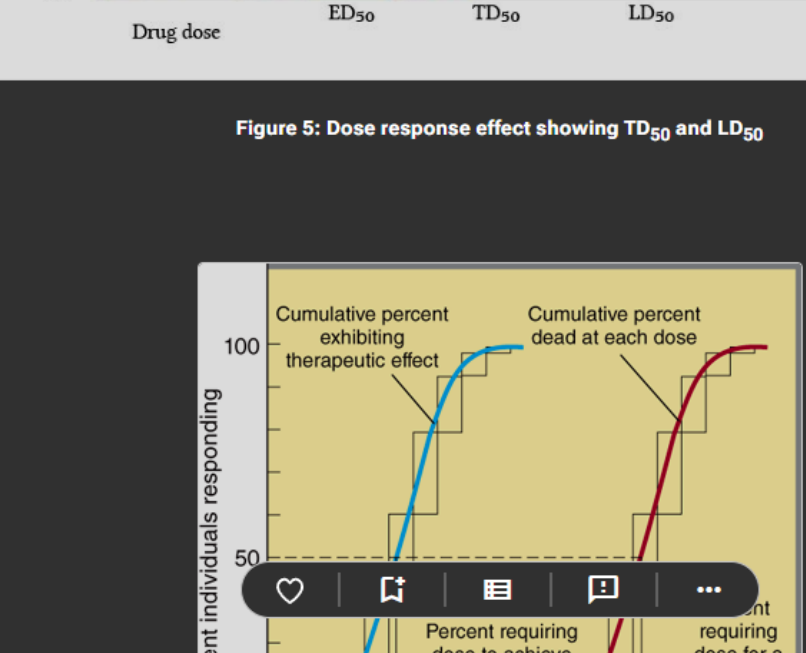


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Figure 6: change plot shaded boxes are the frequency distribution; open boxes are frequency distribution at a particular dose

In Figure 6, when the frequency response of the lethal dose is summated and plotted against log dose, a quantal dose-response curve is generated. This can now be used to calculate the LD50 (red line in Figure 6). Thus, TI indicates how safely separated toxic and therapeutic doses are for a drug.

How Do We Know How Much Drug to Prescribe?

A physician starts a patient on the lowest dose that produces a desired effect because the lowest dose is the least toxic dose. If the desired effect is not achieved, the next step may be to increase the dose. Increasing drug dosage increases drug plasma concentration, which might lead to a stronger effect on the body. However, we do not want to increase the plasma concentration to the point where potential side effects can occur.

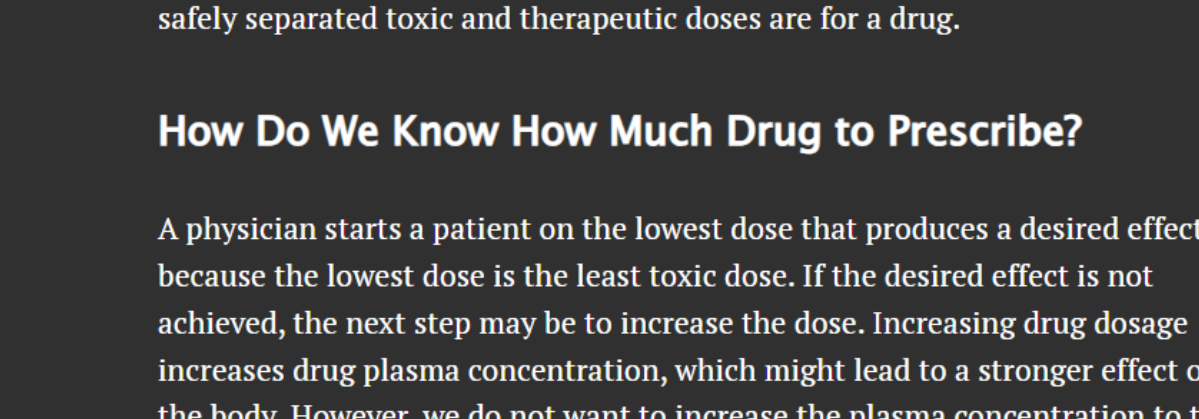

Figure 7

The minimum plasma concentration at which a drug shows clinical benefit is known as the **minimum effective concentration**. The minimum plasma concentration above which a drug may demonstrate toxic effects is termed the **minimum toxic concentration**. The range of concentrations between the minimum effective concentration and minimum toxic concentration is the **therapeutic window** (Figure 7).

What is the therapeutic window?

A given drug should be maintained within this therapeutic window to produce the desired effects and avoid toxic effects. A drug with a narrow therapeutic window (such as the anticoagulant warfarin) must be monitored closely, since the effective concentration and toxic concentration are in proximity. On the other hand, anti-inflammatory drug ibuprofen, has a much wider therapeutic range, so it can be taken in a wide range of doses. The wider therapeutic window lowers the potential for adverse effects. Thus, the therapeutic window describes the safe dosing range for administration in clinical practice.

CLINICAL CORRELATION

Ibuprofen is a nonsteroidal anti-inflammatory drug (NSAID). Although it is generally safe in a range of doses, it should be used with caution by specific groups of people, such as those with gastric ulcers or chronic kidney disease.

Margin of Safety

You may have noticed that the therapeutic index does not take into account the comparative slopes of the two curves. What if the curves aren't perfectly parallel as they are in Figure 4, so the slopes are not the same? Using the median dose, as we do with the therapeutic index, doesn't account for different slopes between the efficacy curves and the toxicity curves. The margin of safety is calculated based on animal studies before drugs are tested in humans. It considers the dose th) and compares it to the dose that is lethal in just 1% of animals (LD$_1$):

$$\text{margin of safety} = LD_1 \div ED_{99}$$

The margin of safety is one of the factors used to determine starting doses in humans when moving from preclinical to clinical drug trials.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to RT, how do you explain the weekly monitoring for warfarin?

You explain to RT that warfarin is a very important medication and is called an anticoagulant. Hence, it is sometimes called a blood thinner. It helps with the blood flow and not letting it coagulate. This is especially important after a valve replacement. But dosing for warfarin is slightly different from other medications she may have taken. This is because if the dose of warfarin is low, then the blood can still coagulate, which can be potentially harmful. On the other hand, if the dose is high, then more anti-coagulation occurs, which can risk bleeding. Hence, the dosing has to be monitored with a small window. You explain that drugs such as warfarin fall under the category of "narrow therapeutic index." Warfarin monitoring is required for patients to ensure that we do not go outside this dosing window to prevent bleeding. This requires that the patient be monitored every week during the initial therapy to establish an appropriate dose for the patient.

Summary

- Drugs can interact with cellular enzymes or receptors, either stimulating or inhibiting their action. This includes drugs that bind directly to a specific enzyme or receptor, as well as physiologic and chemical inhibitors.
- Drug effect describes the downstream consequences that occur after a drug-target interaction. It can be measured through potency (amount of drug required to produce an effect), efficacy (ability of a drug to produce a maximum response), and affinity (how well a drug binds to its receptor).
- The therapeutic window is the range of concentrations between the minimum effective concentration and minimum toxic concentration of a drug. A drug should be maintained within this window to produce desired effects and avoid toxic effects.
- The therapeutic index is a ratio of the median dose that produces a toxic effect to the median dose that produces a desired effect. It is important for safe and effective administration of a drug.
- The margin of safety is the ratio of the dose that is effective in 99% of animals to the dose that is toxic in just 1% of animals. It is one of the factors used to determine starting doses in humans when moving from preclinical to clinical drug trials.

Review Questions

1. If the minimum effective concentration of drug X is 5 mg/mL, and the minimum toxic concentration is 10 mg/mL, which of the following is the size of drug X's th

A. ● 5 mg/mL
B. ○ 10 mg/mL
C. ○ 15 mg/mL
D. ○ 20 mg/mL
E. ○ 25 mg/mL

Hide Explanation

The correct answer is 5 mg/mL (A). The therapeutic window of a drug is any concentration between the minimum effective concentration and the minimum toxic concentration. The therapeutic window for drug X would thus be 5-10 mg/mL, and the size of the therapeutic window would be 10 mg/mL – 5 mg/mL = 5 mg/mL. A value of 10 mg/mL (B) is the minimum toxic concentration, not the size of the therapeutic window. A value of 15 mg/mL (C) adds the minimum effective and minimum toxic concentrations together, which is incorrect and represents a toxic concentration of drug X. Values of 20 mg/mL (D) and 25 mg/mL (E) represent toxic concentrations of drug X, surpassing the minimum toxic concentration of 10 mg/mL.

2. Which of the following pairs of parameters is best used to calculate a drug's margin of safety?

A. ○ ED$_{50}$ and
B. ○ ED$_{50}$ and TD$_{50}$
C. ○ ED$_{50}$ and LD$_{50}$
D. ○ ED$_{99}$ and TD$_{50}$
E. ● ED$_{99}$ and LD$_1$

Hide Explanation

The correct answer is ED$_{99}$ and LD$_1$ (E). Margin of safety is a measure calculated in animals to help determine how to design human clinical trials. Margin of safety is the dose that is lethal in 1% of animals divided by the dose that is effective in 99% of animals. ED$_{50}$ and TD$_1$ (A) is incorrect because this pair describes a drug effective in only 50% of animals and toxic to 1%. ED$_{50}$ and TD$_{50}$ (B) describes the therapeutic index, which is the effective dose for 50% of the population and toxic in 50% of the population. ED$_{50}$ and LD$_{50}$ (C) is incorrect because it describes a drug dose effective in 50% of the population and lethal in 50%. ED$_{99}$ and TD$_{50}$ (D) is incorrect because it describes a drug dose effective in 99% of the population and toxic in 50%.

3. Deferoxamine is an iron chelator that can reduce iron toxicity by binding iron for elimination by the kidneys. Which of the following is the best classification for this drug?

A. ● Chemical
B. ○ Inverse agonist
C. ○ Irreversible antagonist
D. ○ Physiologic inhibitor

Hide Explanation

The correct answer is chemical inhibitor (A). Inverse agonist (B) and irreversible antagonist (C) are incorrect because the chelator does not bind any specific enzyme or drug target. Physiologic inhibitor (D) is incorrect because the drug does not have the opposite effect of any one enzyme or drug receptor.

4. Warfarin dosing is monitored weekly to prevent the bleeding risk because

A. ○ it has a large therapeutic window
B. ● it has a narrow therapeutic index
C. ○ it has a high median lethal dose (LD$_{50}$)
D. ○ it has a high median toxic dose (TD$_{50}$)

Hide Explanation

The correct answer is it has a narrow therapeutic index (B). The risk of bleeding increases for agents with a smaller therapeutic index, as the overlap between the ED50 and TD50 can occur. Choice (A) is incorrect because drugs that have larger therapeutic windows have less risk of reaching doses that can cause adverse effects. Choice (C) is incorrect because a high lethal dose would not necessitate close monitoring. Choice (D) is incorrect because a high toxic dose would not require dose monitoring as the risk of causing harm is lower.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain distinct mechanisms of action of chemical and physiologic antagonists and identify their roles within specific clinical applications.~~
02 ☑ ~~Interpret quantal dose-response data to deduce drug attributes such as TD50, ED50, safety, and determine the therapeutic index.~~
03 ☑ ~~Identify relevant pharmacodynamic and pharmacokinetic principles and apply those to solve clinical problems and predict outcomes.~~

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Objective Confidence: 3/3